Hieu Dinh Nhat Anh Quoc Anh

Computer Vision Course

February 26, 2026

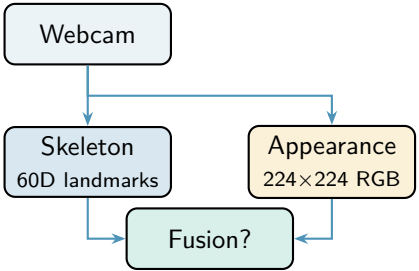
Research Questions & Pipeline

Core Question

Does fusing **skeleton** (geometric) and **appearance** (visual) features from a single RGB camera improve hand gesture recognition?

Q1 (Multi-Modal): How do methods of increasing complexity compare? Does fusion add value?

Q2 (Deployment): What is the *lightest model sufficient* for reliable control in a real-time browser game?



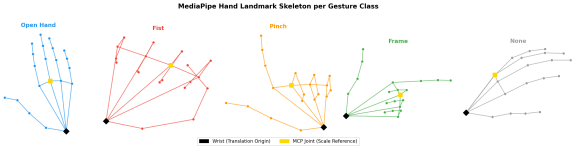
Dataset & Two Feature Streams

HaGRID v2 (Subset)

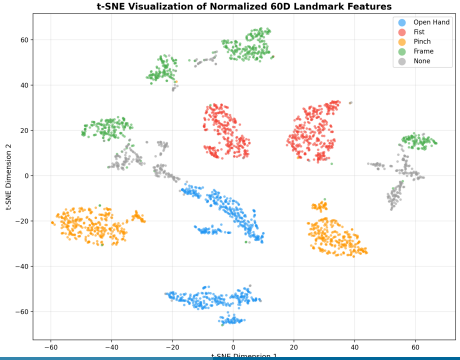
- 23,850 samples, 13,250 subjects
- 5 classes: Pinch, Fist, Open, Frame, None
- **Person-aware** Group 5-Fold CV (no subject leakage)

Two Modalities from One Camera:

- **Skeleton:** MediaPipe → 21 joints → normalized 60D vector
- **Appearance:** YOLO crop → 224×224 RGB



Skeleton: geometrically distinct classes



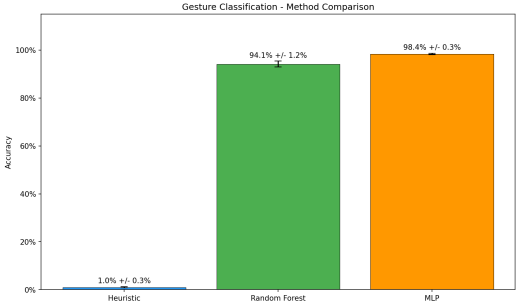
t-SNE:

Methods & Results

Six methods, two modalities:

Method	Modality	Acc	Params	Size
Heuristic	Skeleton	1.0%	∼20	170 B
Random Forest	Skeleton	94.1%	∼100K	8.6 MB
MLP	**Skeleton**	**99.8%**	**14.7K**	**227 KB**
CNN	Appear.	100.0%	1.5M	0.3 MB
YOLOv8n	Appear.	99.5%†	3.0M	6.2 MB
Fusion	**Both**	**100.0%**	**—**	**0.5 MB**

†mAP@50 (detection metric)



Person-aware Group 5-Fold CV

13,250 subjects × 5 folds

= every person tested as "unseen"

Key Finding: The Ceiling Effect

Both modalities **independently** achieve near-perfect accuracy.
Fusion adds **zero measureable benefit**.

Why this is a valid finding:

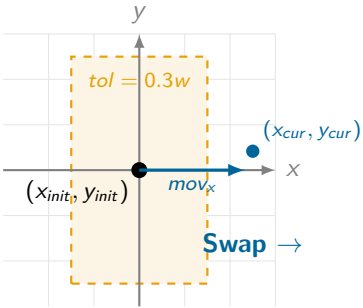
1. 5 gestures are **geometrically unambiguous**
   (fist $\neq$ open $\neq$ pinch)
2. Person-aware CV: **zero identity leakage**
3. 13,250 subjects, clean labels

What this tells us:

1. **Task difficulty** is the dominant factor,
   not model capacity
2. For simple vocabularies:
   single-modality **MLP suffices**
3. Fusion value appears at harder tasks
   (full HaGRID: 18 classes, 95.5% SOTA)

Beyond Classification: Mathematical Interaction Model

99.8% accuracy $\neq$ usable control. Hand jitter $\rightarrow$ erratic cursor. Transient errors $\rightarrow$ dropped tiles.
Solution: Adapt the directional threshold model from Huda et al. 2025 (wheelchair control).



Huda et al.: wheelchair direction.

Ours: puzzle tile swap.

Let $\Delta x = x_{cur} - x_{init}$, $\Delta y = y_{cur} - y_{init}$

Action	Condition
Swap $\rightarrow$	$\Delta x > mov_x$ **AND** $\lvert\Delta y\rvert < tol$
Swap $\leftarrow$	$\Delta x < -mov_x$ **AND** $\lvert\Delta y\rvert < tol$
Swap $\uparrow$	$\Delta y < -mov_y$ **AND** $\lvert\Delta x\rvert < tol$
Swap $\downarrow$	$\Delta y > mov_y$ **AND** $\lvert\Delta x\rvert < tol$

$mov_{x,y} = 0.6 \times \text{tile_size}$
$tol = 0.3 \times \text{tile_size}$

Origin resets after each swap
$\rightarrow$ continuous multi-step swiping.

Conclusion

1. **Multi-modal investigation:** Both skeleton (MLP, 99.8%) and appearance (CNN, 100%) independently saturate accuracy for 5 coarse gestures. Fusion adds no benefit — a **ceiling effect** driven by task simplicity, not model weakness.
2. **Practical deployment:** A **227 KB MLP** on 60D MediaPipe landmarks is sufficient for reliable real-time control. No CNN or fusion needed.
3. **From classification to control:** Adapting the mathematical threshold model from **Huda et al. 2025** (wheelchair) to puzzle tile swiping provides jitter-free interaction through mov_x, mov_y, and tol parameters.
4. **Deployed system:** Runs 100% client-side in any browser at 30+ FPS.

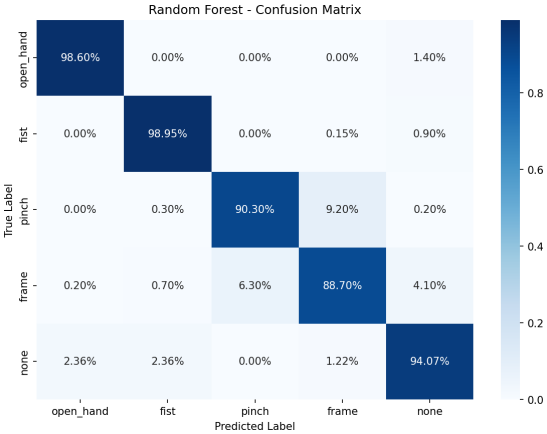
Thank You

Questions & Game Demo

Hieu Dinh · Nhat Anh · Quoc Anh

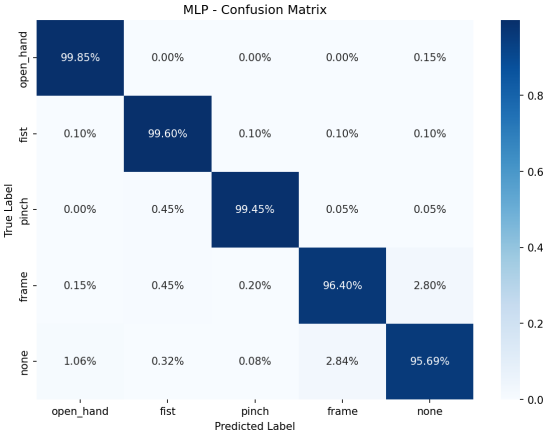
Backup: Confusion Matrices

Random Forest (94%)



Frame ↔ None confusion

MLP (99.8%)



Full 60D resolves it

Backup: YOLO Detection Results

YOLOv8n — simultaneous detection + classification

Metric	**Score**
mAP@50	99.5%
mAP@50:95	90.7%
Parameters	3.0M
Model size	6.2 MB

Different task than classification (includes localization).
Useful when hand bounding box is unknown at inference.

But: 20× larger than MLP — overkill when MediaPipe provides landmarks.

Backup: SOTA Context

HaGRID baselines (full 18-class set, 554K images):

Method	**Params**	**F1**
SSDLite+MobileNetV3-S	—	57.7
SSDLite+MobileNetV3-L	—	71.6
ResNet-18	11M	89.6
ResNeXt-101	44M	87.0
ResNet-152	60M	95.5
Ours: MLP on 5 classes	**14.7K**	**99.8**

Not directly comparable (5 vs 18 classes), but shows:
a tiny MLP saturates a simple subset of the same dataset.